EE6301
25 Bilingual MCQs
25道双语单选模拟卷

17 concepts + 8 calculations · 60 minutes
17道概念 + 8道短算 · 60分钟

Body text / 正文：11 pt
English first, Chinese immediately below. / 英文在前，中文紧随。

2026-10-11 · Course notes, sample explanations and original practice
课件知识、样题参考解析与原创练习

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Original 25-question mock / 原创25题模拟

EN: Single best answer, five options per question, 4 marks each, 60 minutes. Try the English-only view before showing translations or explanations. The bank contains 17 concepts and 8 short calculations, based on the lecturer’ s approximate weighting. Numerical values and option order are original practice choices.

中文： 单个最佳答案，每题五选项，每题4分，60分钟。先用纯英文模式作答，再打开中文与解析。按老师大致比例安排17概念、8短计算；数值与选项顺序为原创练习设定。

Prediction Q1: Optical colour generation in ART / 模拟Q1：ART光学显色原理

EN: In an antigen rapid test (ART), antibodies capture the target and gold nanoparticles accumulate at the test line. Which optical phenomenon mainly produces the visible colour of the gold nanoparticles?

中文： 在抗原快速检测（ART）中，抗体捕获目标，金纳米粒子在测试线处富集。金纳米粒子的可见颜色主要来自哪一种光学现象？

EN: A. Fluorescence from an excited molecular dye

中文： A. 受激分子染料的荧光

EN: B. Localized surface plasmon resonance (LSPR)

中文： B. 局域表面等离激元共振（LSPR）

EN: C. Inelastic Raman scattering by the antigen

中文： C. 抗原的非弹性拉曼散射

EN: D. Total internal reflection at a glass–air interface

中文： D. 玻璃—空气界面的全反射

EN: E. Phosphorescence from a long-lived triplet state

中文： E. 长寿命三重态的磷光

Prediction Q2: Labeled versus label-free sensing / 模拟Q2：有标记与无标记

EN: Assay X measures fluorescence from a dye attached to a detection antibody. Assay Y uses immobilized capture antibodies and measures a resonance-wavelength shift caused by target binding, without adding a reporter dye to the target or detection reagent. Which statement correctly classifies the two assays?

中文： 检测X测量连接在检测抗体上的染料荧光。检测Y使用固定的捕获抗体，通过目标结合引起的共振波长位移读出，且不向目标或检测试剂添加报告染料。以下哪项正确分类了这两种检测？

EN: A. Both are labeled because both use antibodies.

中文： A. 两者都是有标记检测，因为两者都使用抗体。

EN: B. Both are label-free because both detect target binding.

中文： B. 两者都是无标记检测，因为两者都检测目标结合。

EN: C. X is label-free, whereas Y is labeled.

中文： C. X是无标记，Y是有标记。

EN: D. X is labeled, whereas Y is label-free.

中文： D. X是有标记，Y是无标记。

EN: E. Y cannot be label-free because an optical light source is used.

中文： E. Y使用光源，因此不可能是无标记检测。

Prediction Q3: Raman versus fluorescence / 模拟Q3：拉曼与荧光机制

EN: Which statement correctly distinguishes Raman scattering from ordinary fluorescence in the lecture models?

中文： 按课程中的模型，哪项正确区分了拉曼散射与普通荧光?

EN: A. Raman scattering exchanges energy with molecular vibrations via a virtual-state process; fluorescence follows absorption into a real excited state and subsequent radiative emission.

中文： A. 拉曼散射通过虚态过程与分子振动交换能量；荧光先吸收进入真实激发态，再进行辐射发射。

EN: B. Raman scattering is elastic, so the scattered photon must always have the incident photon energy.

中文： B. 拉曼散射是弹性散射，因此散射光子能量必须始终与入射光子相同。

EN: C. Fluorescence requires no absorption of excitation light.

中文： C. 荧光不需要吸收激发光。

EN: D. Every red-shifted optical signal is fluorescence, so Stokes Raman scattering is a form of fluorescence.

中文： D. 所有红移光信号都是荧光，因此斯托克斯拉曼散射属于荧光。

EN: E. Raman scattering and fluorescence must have identical spectra because both can emit lower-energy photons.

中文： E. 拉曼与荧光都可能发射较低能量光子，因此二者的光谱必须相同。

Prediction Q4: Grating density and spectral resolution / 模拟Q4：光栅刻线密度与光谱分辨率

EN: Two ideal diffraction gratings are compared at the same wavelength and diffraction order. They have the same illuminated width, and other instrument limitations are neglected. Grating X has 300 lines mm^{-1}; grating Y has 1200 lines mm^{-1}. Which grating can resolve a smaller wavelength separation, and why?

> **中文：** 在相同波长、相同衍射级次下比较两块理想衍射光栅。它们的受照宽度相同，忽略其他仪器限制。光栅X有300条/mm，光栅Y有1200条/mm。哪块光栅能分辨更小的波长间隔，原因是什么？

EN: A. X, because fewer illuminated grooves always give a larger resolving power.

> **中文：** A. X，因为受照刻线越少，分辨本领总是越大。

EN: B. X, because a lower groove density makes all spectral peaks intrinsically narrower.

> **中文：** B. X，因为低刻线密度会让所有光谱峰本征变窄。

EN: C. They must have the same resolution because the detector has the same number of pixels.

> **中文：** C. 它们的分辨率必须相同，因为探测器像素数相同。

EN: D. Y, solely because the numerical value 1200 is a microscope magnification.

> **中文：** D. Y，唯一原因是数值1200代表显微镜放大倍数。

EN: E. Y, because the higher groove density gives more illuminated grooves over the same width.

> **中文：** E. Y，因为同一受照宽度内，较高刻线密度意味着更多受照刻线。

Prediction Q5: Sensitivity versus limit of detection / 模拟Q5：灵敏度不等于检测限

EN: Sensor B has a steeper linear calibration slope than sensor A, but its blank measurements are also noisier. Can the slope information alone establish that B has a lower limit of detection (LOD), using the course model LOD = 3σ/slope?

中文： 传感器B的线性校准斜率比A更大，但其空白测量噪声也更大。按课程模型LOD=3σ/斜率，仅凭斜率信息能否确定B的检测限更低?

EN: A. Yes; a larger slope always guarantees a lower LOD, regardless of noise.

中文： A. 能；无论噪声怎样，斜率更大都保证LOD更低。

EN: B. Yes; slope and LOD are two names for the same quantity.

中文： B. 能；斜率和LOD是同一个量的两种名称。

EN: C. No; LOD depends on the ratio of blank standard deviation to slope.

中文： C. 不能；LOD取决于空白标准差与斜率之比。

EN: D. No; LOD depends only on selectivity and never on slope.

中文： D. 不能；LOD仅取决于选择性，与斜率永远无关。

EN: E. Yes; greater noise improves the distinction between blank and target signals.

中文： E. 能；噪声更大能改善空白与目标信号的区分。

Prediction Q6: Beer–Lambert short calculation / 模拟Q6：比尔朗伯短算

EN: At a fixed wavelength, a homogeneous sample transmits 25% of the incident intensity through a 1.0 cm optical path. Its concentration and molar extinction coefficient remain unchanged. Under Beer–Lambert's law, what is the transmittance through a 2.0 cm path? Ignore other losses.

> **中文：** 在固定波长下，均匀样本经过1.0 cm光程后的透射强度为入射强度的25%。浓度与摩尔消光系数保持不变。按比尔朗伯定律，光程改为2.0 cm后透过率是多少？忽略其他损耗。

EN: A. 12.5%

> **中文：** A. 12.5%

EN: B. 25.0%

> **中文：** B. 25.0%

EN: C. 50.0%

> **中文：** C. 50.0%

EN: D. 6.25%

> **中文：** D. 6.25%

EN: E. 75.0%

> **中文：** E. 75.0%

Prediction Q7: Radiative rate from quantum yield and lifetime / 模拟Q7：由量子产率与寿命求辐射速率

EN: A fluorophore has fluorescence quantum yield $\Phi = 0.60$ and a single-exponential fluorescence lifetime $\tau = 12$ ns. Using $\Phi = k_r\tau$, what is its radiative decay rate k_r?

> **中文：** 某荧光团的荧光量子产率$\Phi=0.60$，单指数荧光寿命$\tau=12$ ns。使用$\Phi=k_r\tau$，辐射衰减速率k_r是多少?

EN: A. 0.050 ns^{-1}

> **中文：** A. 0.050 ns^{-1}

EN: B. 0.0833 ns^{-1}

> **中文：** B. 0.0833 ns^{-1}

EN: C. 0.20 ns^{-1}

> **中文：** C. 0.20 ns^{-1}

EN: D. 5.0 ns^{-1}

> **中文：** D. 5.0 ns^{-1}

EN: E. 20 ns^{-1}

> **中文：** E. 20 ns^{-1}

Prediction Q8: Given-formula grating resolving power / 模拟Q8：光栅分辨本领给式短算

EN: For an ideal diffraction grating, the resolving power is given by R = mN, where m is the diffraction order and N is the number of illuminated grooves. If N = 2500 and the spectrum is observed in the second order (m = 2), what is R?

> **中文：** 理想衍射光栅的分辨本领在题中给定为R=mN，其中m为衍射级次，N为受照刻线数。如果N=2500，并在第二级衍射（m=2）观察光谱，R是多少?

EN: A. 1250

> **中文：** A. 1250

EN: B. 2500

> **中文：** B. 2500

EN: C. 5000

> **中文：** C. 5000

EN: D. 7500

> **中文：** D. 7500

EN: E. 10000

> **中文：** E. 10000

Prediction Q9: LOD for a continuous concentration / 模拟Q9：连续浓度检测限

EN: A sensor has a linear calibration slope of 0.80 μA mM^{-1}. The standard deviation of the blank current is 0.12 μA. Using LOD = 3σ/slope, what is the concentration LOD? Concentration is continuous; do not round it up to an integer.

> **中文：** 某传感器的线性校准斜率为0.80 μA/mM，空白电流标准差为0.12 μA。使用LOD=3σ/斜率，浓度检测限是多少？浓度为连续量，不要向上取整为整数。

EN: A. 0.15 mM

> **中文：** A. 0.15 mM

EN: B. 0.30 mM

> **中文：** B. 0.30 mM

EN: C. 0.36 mM

> **中文：** C. 0.36 mM

EN: D. 0.40 mM

> **中文：** D. 0.40 mM

EN: E. 0.45 mM

> **中文：** E. 0.45 mM

Prediction Q10: Numerical aperture versus magnification / 模拟Q10：数值孔径与放大倍率

EN: Five microscope objectives are used with the same illumination wavelength. Assume adequate detector sampling and compare their diffraction-limited lateral resolution. Which objective can resolve the smallest separation between two points?

> **中文：** 五个显微物镜使用相同照明波长。假设探测器采样充分，比较衍射极限下的横向分辨能力。哪一个物镜能够分辨两点间最小的距离？

EN: A. Magnification 100×; numerical aperture 0.80

> **中文：** A. 放大倍率 100×；数值孔径 0.80

EN: B. Magnification 40×; numerical aperture 0.95

> **中文：** B. 放大倍率 40×；数值孔径 0.95

EN: C. Magnification 60×; numerical aperture 1.20

> **中文：** C. 放大倍率 60×；数值孔径 1.20

EN: D. Magnification 40×; numerical aperture 1.30

> **中文：** D. 放大倍率 40×；数值孔径 1.30

EN: E. Magnification 100×; numerical aperture 1.25

> **中文：** E. 放大倍率 100×；数值孔径 1.25

Prediction Q11: OCT coherence/interference mechanism / 模拟Q11：OCT 相干干涉机制

EN: Which statement best describes the core measurement principle of optical coherence tomography (OCT)?

中文： 哪项最准确地描述光学相干断层成像（OCT）的核心测量原理?

EN: A. Fluorescent labels emit light after absorption, and only the emission intensity is measured.

中文： A. 荧光标记吸收光后发光，只测量其发射强度。

EN: B. Backreflected or backscattered sample light interferes with light from a reference arm to obtain depth-resolved information.

中文： B. 样品返回的反射或后向散射光与参考臂的光干涉，获取可区分深度的信息。

EN: C. Total internal reflection alone determines the brightness of every tissue layer.

中文： C. 仅靠全反射决定每一层组织的亮度。

EN: D. A piezoelectric transducer emits and receives sound, and no optical source is used.

中文： D. 压电换能器发射和接收声波，不使用光源。

EN: E. One X-ray projection directly reveals the depth of every tissue layer without reconstruction.

中文： E. 一张 X-ray 投影无需重建就直接显示所有组织层的深度。

Prediction Q12: Ultrasound round-trip echo time / 模拟Q12：超声往返回波时间

EN: An ultrasound pulse travels through soft tissue at 1540 m/s. An echo from a single interface returns 40 μs after the pulse is emitted. Assuming a straight path and constant speed, how deep is the interface?

> **中文：** 一束超声脉冲在软组织中以 1540 m/s 传播。脉冲发出后 40 μs 收到某单一界面的回波。假设直线路径和恒定声速，该界面的深度是多少？

EN: A. 0.308 cm

> **中文：** A. 0.308 cm

EN: B. 1.54 cm

> **中文：** B. 1.54 cm

EN: C. 3.08 cm

> **中文：** C. 3.08 cm

EN: D. 6.16 cm

> **中文：** D. 6.16 cm

EN: E. 30.8 cm

> **中文：** E. 30.8 cm

Prediction Q13: Photoacoustic optical-to-acoustic conversion / 模拟Q13：光声的光—声转换

EN: Which sequence correctly describes signal generation and detection in photoacoustic imaging?

> **中文：** 哪一信号产生与检测顺序正确描述光声成像?

EN: A. Ultrasound absorption → fluorescent emission → optical detector

> **中文：** A. 超声吸收→荧光发射→光探测器

EN: B. Optical excitation → total internal reflection → acoustic generation without heating

> **中文：** B. 光激发→全反射→无加热的声产生

EN: C. X-ray absorption → nuclear decay → piezoelectric emission

> **中文：** C. X-ray 吸收→核衰变→压电发射

EN: D. Electrical current in DNA → magnetic field → fluorescence detector

> **中文：** D. DNA 中的电流→磁场→荧光探测器

EN: E. Optical absorption → transient heating and thermal expansion → acoustic wave → ultrasound transducer

> **中文：** E. 光吸收→瞬态加热与热膨胀→声波→超声换能器

Prediction Q14: CCD versus CMOS readout / 模拟Q14：CCD 与 CMOS 读出

EN: Which statement correctly distinguishes the CCD and CMOS detectors described in the slides?

中文： 哪项正确区分课件中的 CCD 与 CMOS 探测器?

EN: A. A CCD is defined by conversion and amplification at every pixel, while CMOS transfers all charge to one output node.

中文： A. CCD 的定义是在每个像素进行转换和放大，而 CMOS 把全部电荷搬到单一输出节点。

EN: B. Both CCD and CMOS detectors measure only acoustic pressure and cannot detect light.

中文： B. CCD 与 CMOS 都只测声压，不能探测光。

EN: C. A CCD transfers pixel charge towards an output node, whereas CMOS integrates conversion/amplification circuitry within pixels.

中文： C. CCD 把像素电荷向输出节点转移，而 CMOS 在像素内整合转换/放大电路。

EN: D. A CMOS detector necessarily has zero electronic noise under all operating conditions.

中文： D. CMOS 探测器在任何工作条件下电子噪声必定为零。

EN: E. A CCD increases optical numerical aperture merely by transferring charge.

中文： E. CCD 仅靠电荷转移就能增大光学数值孔径。

Prediction Q15: Amperometric versus potentiometric sensing / 模拟Q15：电流型与电位型传感

EN: A biosensor is operated at a selected electrode potential, and the current caused by an electrochemical reaction is recorded as the analytical signal. Which sensing mode is this?

> **中文：** 某生物传感器在选定的电极电位下工作，以电化学反应产生的电流作为分析信号记录。这属于哪种传感模式?

EN: A. Amperometric sensing

> **中文：** A. 电流型传感（amperometric）

EN: B. Potentiometric sensing under approximately open-circuit conditions

> **中文：** B. 近似开路条件下的电位型传感（potentiometric）

EN: C. A hyperspectral measurement of each pixel's optical spectrum

> **中文：** C. 测量每个像素光学光谱的高光谱传感

EN: D. A measurement of a fluorescence-emission wavelength only

> **中文：** D. 仅测荧光发射波长的传感

EN: E. A pure optical time-of-flight measurement without electrodes

> **中文：** E. 不使用电极的纯光学飞行时间测量

Prediction Q16: Potentiometric pH and ion selectivity / 模拟Q16：电位型 pH 与离子选择性

EN: A conventional ion-selective pH electrode measures a potential difference relative to a stable reference electrode, with the response related to hydrogen-ion activity. Which statement is correct?

> **中文：** 常规离子选择性 pH 电极相对稳定的参考电极测量电位差，其响应与氢离子活度有关。哪项正确？

EN: A. It must be classified as amperometric merely because electrodes are present.

> **中文：** A. 仅因使用了电极，就必须分类为电流型传感。

EN: B. It measures the density of an optical diffraction grating.

> **中文：** B. 它测量光学衍射光栅的密度。

EN: C. It detects DNA sequences by ionic-current blockades in a nanopore.

> **中文：** C. 它通过纳米孔离子电流阻塞检测 DNA 序列。

EN: D. It is an example of potentiometric sensing; the pH-sensitive interface supplies ion selectivity.

> **中文：** D. 它是电位型传感的实例；pH 敏感界面提供离子选择性。

EN: E. Its reference electrode should change arbitrarily with the analyte to cancel all pH sensitivity.

> **中文：** E. 参考电极应随分析物任意变化，以抵消全部 pH 敏感性。

Prediction Q17: ECG / EEG / EMG: terminology and signal sources / 模拟Q17：ECG / EEG / EMG的英文名词与信号来源

EN: Which set correctly matches the three biopotential recordings to their main biological sources?

中文： 哪一组正确对应了三种生物电记录及其主要生物来源?

EN: A. ECG: brain; EEG: heart; EMG: skeletal muscle.

中文： A. ECG：脑；EEG：心脏；EMG：骨骼肌。

EN: B. ECG: heart; EEG: brain; EMG: skeletal muscle.

中文： B. ECG：心脏；EEG：脑；EMG：骨骼肌。

EN: C. ECG: heart; EEG: skeletal muscle; EMG: brain.

中文： C. ECG：心脏；EEG：骨骼肌；EMG：脑。

EN: D. ECG: skeletal muscle; EEG: brain; EMG: heart.

中文： D. ECG：骨骼肌；EEG：脑；EMG：心脏。

EN: E. ECG: brain; EEG: skeletal muscle; EMG: heart.

中文： E. ECG：脑；EEG：骨骼肌；EMG：心脏。

Prediction Q18: Action potentials and depolarization / 模拟Q18：动作电位与去极化

EN: In the neuronal action-potential model discussed in the course, which event best describes the rising depolarization phase after threshold is reached?

中文： 在课程讨论的神经元动作电位模型中，达到阈值后的上升去极化阶段最符合哪种过程?

EN: A. K+ leaves the cell, making the membrane potential return toward its resting value.

中文： A. K+流出细胞，使膜电位向静息值恢复。

EN: B. The membrane becomes more negative than its resting value.

中文： B. 膜电位变得比静息值更负。

EN: C. All ion channels remain closed, so the membrane potential stays unchanged.

中文： C. 所有离子通道都保持关闭，因此膜电位不变。

EN: D. Voltage-gated Na+ channels open, and Na+ influx makes the cell interior less negative.

中文： D. 电压门控Na+通道打开，Na+内流使细胞内部变得不那么负。

EN: E. The resting potential is zero because no action potential is present.

中文： E. 没有动作电位时，静息电位为零。

Prediction Q19: Patch clamp versus multielectrode arrays / 模拟Q19：膜片钳与多电极阵列

EN: Which comparison between patch clamp and a multielectrode array (MEA) is correct for the recording methods shown in the course?

中文： 对于课程展示的记录方法，膜片钳（patch clamp）与多电极阵列（MEA）的哪项比较正确？

EN: A. Patch clamp uses a glass micropipette–membrane interface; an MEA uses multiple electrode sites to record electrical activity at multiple locations.

中文： A. 膜片钳使用玻璃微吸管与细胞膜的接触界面；MEA使用多个电极位点在多个位置记录电活动。

EN: B. Patch clamp records only light intensity; an MEA records only acoustic pressure.

中文： B. 膜片钳只记录光强；MEA只记录声压。

EN: C. Patch clamp uses a planar array of many electrodes, whereas an MEA uses one glass micropipette sealed to the membrane.

中文： C. 膜片钳使用许多电极组成的平面阵列，而MEA使用一根密封在细胞膜上的玻璃微吸管。

EN: D. An MEA necessarily provides a separate intracellular recording from every individual cell in the culture.

中文： D. MEA必然能分别对培养体系中的每个细胞进行胞内记录。

EN: E. Patch clamp and an MEA are filters whose only function is to remove low-frequency interference.

中文： E. 膜片钳与MEA都是滤波器，唯一作用是去除低频干扰。

Prediction Q20: Differential measurement and common-mode interference / 模拟Q20：差分测量与共模干扰

EN: Assume an ideal differential amplifier receives V1 = s1 + n and V2 = s2 + n, where exactly the same interference n appears at both inputs. Which statement is correct?

> **中文：** 假定理想差分放大器的输入为V1 = s1 + n、V2 = s2 + n，其中完全相同的干扰n出现在两个输入端。哪项正确?

EN: A. The differential input is s1 + s2 + 2n.

> **中文：** A. 差分输入为s1 + s2 + 2n。

EN: B. The differential input is always zero, even if s1 differs from s2.

> **中文：** B. 即使s1与s2不同，差分输入也总为零。

EN: C. The interference remains unchanged because subtraction cannot affect noise.

> **中文：** C. 干扰完全不变，因为相减不可能影响噪声。

EN: D. Subtraction removes every possible disturbance, even if it differs between the two electrodes.

> **中文：** D. 相减会去除所有可能的干扰，即使两个电极上的干扰不同。

EN: E. The differential input is s1 − s2; the identical common-mode term n cancels.

> **中文：** E. 差分输入为s1 − s2；相同的共模项n被抵消。

Prediction Q21: Input and output impedances of biopotential amplifiers / 模拟Q21：生物电位放大器的输入与输出阻抗

EN: Which impedance combination is most appropriate for a biopotential amplifier intended to measure a weak voltage signal with minimal loading and then drive an external load?

中文： 生物电位放大器需要尽量减少对微弱电压信号的加载，并驱动外部负载；哪种阻抗组合最合适？

EN: A. Low input impedance and high output impedance.

中文： A. 低输入阻抗与高输出阻抗。

EN: B. Low input impedance and low output impedance.

中文： B. 低输入阻抗与低输出阻抗。

EN: C. High input impedance and low output impedance.

中文： C. 高输入阻抗与低输出阻抗。

EN: D. High input impedance and high output impedance.

中文： D. 高输入阻抗与高输出阻抗。

EN: E. Zero input impedance, so the electrode signal is short-circuited.

中文： E. 零输入阻抗，从而把电极信号短路。

Prediction Q22: Filter passbands and stopbands / 模拟Q22：滤波器的通带与阻带

EN: A filter attenuates frequencies below 10 Hz and above 100 Hz while passing the intermediate frequency band. Which filter type is described?

中文： 某滤波器衰减低于10 Hz和高于100 Hz的成分，同时通过中间频带；这描述了哪类滤波器？

EN: A. Low-pass filter.

中文： A. 低通滤波器。

EN: B. Band-pass filter.

中文： B. 带通滤波器。

EN: C. High-pass filter.

中文： C. 高通滤波器。

EN: D. Band-reject filter.

中文： D. 带阻滤波器。

EN: E. All-pass filter.

中文： E. 全通滤波器。

Prediction Q23: A short differential-voltage and gain calculation / 模拟Q23：差分电压与放大增益的简单计算

EN: An ideal differential amplifier has Vout = G(V1 − V2). If V1 = 1.30 mV, V2 = 1.10 mV and G = 1000, what is Vout?

> **中文：** 理想差分放大器满足Vout = G(V1 − V2)。若V1 = 1.30 mV、V2 = 1.10 mV、G = 1000，Vout是多少?

EN: A. 0.20 mV.

> **中文：** A. 0.20 mV。

EN: B. 1.20 V.

> **中文：** B. 1.20 V。

EN: C. 0.40 V.

> **中文：** C. 0.40 V。

EN: D. 0.20 V.

> **中文：** D. 0.20 V。

EN: E. 2.40 V.

> **中文：** E. 2.40 V。

Prediction Q24: Series contact resistances, current and unit conversion / 模拟Q24：串联接触电阻、电流与单位换算

EN: In a simplified test circuit, model the two electrode contacts as pure resistors of 2.0 kΩ and 3.0 kΩ connected in series, and ignore all other circuit elements. If 10 mV is applied across the pair, what current flows through the series path?

> **中文：** 在简化测试电路中，将两个电极接触分别建模为2.0 kΩ与3.0 kΩ的纯电阻并串联，忽略所有其他电路元件。若两者整体承受10 mV电压，串联通路中的电流是多少？

EN: A. 2.0 mA.

> **中文：** A. 2.0 mA。

EN: B. 5.0 µA.

> **中文：** B. 5.0 µA。

EN: C. 2.0 µA.

> **中文：** C. 2.0 µA。

EN: D. 3.33 µA.

> **中文：** D. 3.33 µA。

EN: E. 10.0 µA.

> **中文：** E. 10.0 µA。

Prediction Q25: Linear-region LOD and an integer detection threshold / 模拟Q25：线性区、LOD与整数检测阈值

EN: Within its valid linear region, a capacitive virus sensor gives 100 pF at 20 viruses and 700 pF at 80 viruses. Assume that the same linear calibration slope remains valid near the detection threshold. The output standard deviation is 14 pF. Use LOD = 3σ/m, where m is the calibration slope. If the answer must be a whole-virus count at or above the calculated threshold, what is the smallest detectable integer count?

中文： 在有效线性区内，电容式病毒传感器在20个病毒时输出100 pF，在80个病毒时输出700 pF。假定同一线性校准斜率在检测阈值附近仍然有效。输出标准差为14 pF。采用LOD = 3σ/m，其中m为校准斜率。若答案必须是达到或超过计算阈值的完整病毒计数，最小可检出的整数计数是多少？

EN: A. 4 viruses.

中文： A. 4个病毒。

EN: B. 5 viruses.

中文： B. 5个病毒。

EN: C. 3 viruses.

中文： C. 3个病毒。

EN: D. 2 viruses.

中文： D. 2个病毒。

EN: E. 4.2 viruses.

中文： E. 4.2个病毒。

Answer key / 答案速查

Q1: B · Q2: D · Q3: A · Q4: E · Q5: C

Q6: D · Q7: A · Q8: C · Q9: E · Q10: D

Q11: B · Q12: C · Q13: E · Q14: C · Q15: A

Q16: D · Q17: B · Q18: D · Q19: A · Q20: E

Q21: C · Q22: B · Q23: D · Q24: C · Q25: B

Explanations and sources / 双语解析与来源

Prediction Q1: Answer B / 模拟Q1：答案B

EN: Gold nanoparticles have a visible optical response associated with LSPR. Antibody binding determines biological recognition, whereas nanoparticle accumulation and LSPR provide the coloured readout. A and E describe molecular luminescence, C describes vibrational Raman scattering, and D describes a ray-guiding condition; none is the main colour mechanism in this gold-nanoparticle ART.

中文： 金纳米粒子的可见光学响应与LSPR有关。抗体结合负责生物识别，粒子富集及LSPR提供色线读出。A、E是分子发光过程，C是振动拉曼散射，D是导光的全反射条件；它们都不是本题金纳米粒子ART的主要显色机制。

Source / 来源：W3 PDF p76；T3-0106。

Prediction Q2: Answer D / 模拟Q2：答案D

EN: A fluorescent reporter dye makes X a labeled assay. Y reads the binding-induced optical change directly, so it is label-free under the stated conditions. A capture antibody is a recognition element, not automatically a reporter label. Label-free sensing can still require illumination; the term does not mean 'without a light source'.

中文： X借助荧光报告染料，因此是有标记检测。Y在题给条件下直接读结合引起的光学变化，因此是无标记检测。捕获抗体是识别元件，不能自动当作报告标签；无标记传感仍可以需要照明，它不等于“没有光源”。

Source / 来源：W2 PDF p89；W3 PDF p13–18；T3-0016–0020。

Prediction Q3: Answer A / 模拟Q3：答案A

EN: Raman is inelastic scattering associated with vibrational energy exchange and a virtual-state description. Fluorescence involves a real electronic excited state, relaxation and radiative return. A lower-energy or longer-wavelength output alone does not identify the process: both Stokes Raman scattering and ordinary Stokes fluorescence can be red-shifted. B confuses Raman with elastic scattering; C omits absorption; D and E confuse a shared spectral direction with an identical mechanism.

中文： 拉曼是与振动能量交换有关的非弹性散射，用虚态描述。荧光涉及真实电子激发态、弛豫和辐射回到低能态。仅凭输出能量更低或波长更长，不能判断机制：斯托克斯拉曼及普通斯托克斯荧光都可能红移。B把拉曼混成弹性散射，C漏掉吸收，D、E把相同的光谱方向误当相同的机制。

Source / 来源：W2 PDF p60–62、p74–75；T2-0070–0078、T2-0088–0096。

Prediction Q4: Answer E / 模拟Q4：答案E

EN: At equal illuminated width, Y has more illuminated grooves and therefore a greater ideal grating resolving power at the same order. Greater resolving power means a smaller resolvable wavelength difference. The same detector pixel count does not by itself force equal spectral resolution. Groove density is neither microscope magnification nor a universal claim about intrinsic source linewidth. The stated conditions matter: a real spectrometer may also be limited by slit width, aberrations or detector sampling.

中文： 受照宽度相同时，Y有更多受照刻线，因此在同级次下具有更高的理想光栅分辨本领；更高分辨本领对应更小可分辨波长差。像素数相同本身不保证光谱分辨率相同。刻线密度既不是显微镜放大倍数，也不保证光源本征线宽改变。题给条件很关键：真实光谱仪还可能受狭缝、像差或探测器采样限制。

Source / 来源：T5-0059、T5-0061–0063（300/1200密度比较及quiz承诺）；W2 PDF p35仅为光谱仪结构背景，原PDF未找到该数字比较图页。

Prediction Q5: Answer C / 模拟Q5：答案C

EN: Sensitivity is the calibration slope, whereas LOD is a concentration threshold estimated from 3σ/slope. Increasing slope tends to reduce LOD, but increasing blank noise tends to raise it. Without their relative changes, the net effect cannot be determined. This question tests the model conceptually rather than requiring a numerical calculation.

中文： 灵敏度是校准斜率，而LOD是由3σ/斜率估计的浓度阈值。斜率增大倾向于降低LOD，空白噪声增大倾向于提高LOD；不知道二者相对变化，就不能确定最终LOD。此题检查模型理解，不要求数值计算。

Source / 来源：W1 PDF p59–62；T1-0053–0058。

Prediction Q6: Answer D / 模拟Q6：答案D

EN: Beer–Lambert's law gives A = εCL = −log10(T). Doubling L doubles absorbance, so T_2 = 10^(−2A_1) = T_1^2 = 0.25^2 = 0.0625 = 6.25%. Transmittance does not decrease linearly as 25%/2; absorbance is the quantity linear in path length under the stated conditions.

中文： 比尔朗伯定律A=εCL=−log10(T)。光程加倍使吸光度加倍，因此 T_2=10^(−2A_1)=T_1^2=0.25^2=0.0625=6.25%。透过率不是按25%÷2线性下降；在题给条件下，随光程线性变化的是吸光度。

Source / 来源：W2 PDF p37–38、p41–43；T2-0041–0051。

Prediction Q7: Answer A / 模拟Q7：答案A

EN: k_r = Φ/τ = 0.60/(12 ns) = 0.050 ns^{-1} = 5.0 × 10^7 s^{-1}. The total decay rate is 1/τ ≈ 0.0833 ns^{-1}, so B is the total rate, not the radiative rate. Quantum yield is dimensionless, and the resulting rate has inverse-time units.

中文： k_r=Φ/τ=0.60/(12 ns)=0.050 ns^{-1}=5.0×10^7 s^{-1}。总衰减速率是1/τ≈0.0833 ns^{-1}，因此B给的是总速率，不是辐射速率。量子产率无量纲，所得速率单位为时间的倒数。

Source / 来源：W2 PDF p83、p85、p87；T2-0105–0107。

Prediction Q8: Answer C / 模拟Q8：答案C

EN: Use the supplied formula: R = mN = 2 × 2500 = 5000. Resolving power is dimensionless. N is the total number of illuminated grooves, not the groove density by itself. If groove density were supplied instead, the illuminated width would also be needed to determine N. This is a given-formula calculation, not a claim that the lecture required memorizing R = mN.

中文： 使用题给公式：R=mN=2×2500=5000，分辨本领无量纲。N是受照刻线总数，不是刻线密度本身。如果题目只给刻线密度，还需要受照宽度才能求N。本题属于给式代入，不声称老师要求背R=mN。

Source / 来源：T5-0061–0063（刻线密度及参与刻线数与分辨率的关系）；W2 PDF p35仅为光谱仪背景。原PDF未定位R=mN公式页，本题将公式直接给出。

Prediction Q9: Answer E / 模拟Q9：答案E

EN: LOD = 3(0.12 μA)/(0.80 μA mM^{-1}) = 0.45 mM. The current units cancel, leaving concentration. A omits the factor of three, while C is the threshold current 3σ = 0.36 μA before division by slope, not the concentration LOD. Because the requested quantity is continuous concentration, no integer ceiling is applied.

中文： LOD=3×0.12 μA/(0.80 μA/mM)=0.45 mM。电流单位抵消，最终留下浓度单位。A漏乘3；C是除以斜率前的阈值电流3σ=0.36 μA，不是浓度LOD。题求连续浓度，因此不做整数向上取整。

Source / 来源：W1 PDF p60–62；T1-0057–0058。

Prediction Q10: Answer D / 模拟Q10：答案D

EN: For the same wavelength, the Rayleigh separation is Δd≈0.61λ/NA. The largest NA, 1.30, gives the smallest resolvable separation. Magnification alone enlarges the image without improving this optical resolution. These numerical choices are original practice options; the recording did not preserve all options of the in-class question.

中文： 相同波长下 Rayleigh 间距为 Δd≈0.61λ/NA。最大 NA=1.30 对应最小可分辨间距。仅提高放大倍率只是放大图像，不改善该光学分辨率。以上数值选项是原创练习，录音未完整保留课堂原题的所有选项。

Source / 来源：Week 4 EE6301 Full Slides (AY2627)(1).pdf, physical PDF p29–30; Week 5 classroom-question review

Prediction Q11: Answer B / 模拟Q11：答案B

EN: OCT compares returning sample light with reference-arm light through interference. Depth information and contrast depend on optical path differences and tissue backreflection/backscattering. It does not require fluorescent labels, and its image intensity should not be identified with total internal reflection.

中文： OCT 通过干涉比较样品返回光与参考臂光。深度信息和对比与光程差、组织反射/后向散射有关。它不要求荧光标记，也不应把图像强度等同全反射强弱。

Source / 来源：Week 4 EE6301 Full Slides (AY2627)(1).pdf, physical PDF p71–79 and p85–86

Prediction Q12: Answer C / 模拟Q12：答案C

EN: The delay includes the outward and return trips, so $d=ct/2=1540\times(40\times10^{-6})/2=0.0308$ m=3.08 cm. The 6.16 cm value is the total travel distance, not the interface depth.

> **中文：** 延迟包含去程与回程，因此 $d=ct/2=1540\times(40\times10^{-6})/2=0.0308$ m=3.08 cm。6.16 cm 是总传播距离，不是界面深度。

Source / 来源：Week 5 EE6301 Full Slides (AY2627).pdf, physical PDF p11 and p13

Prediction Q13: Answer E / 模拟Q13：答案E

EN: Photoacoustic imaging uses light to excite absorbing tissue and detects the acoustic waves produced by heating and thermal expansion. Its input is optical and its measured output is acoustic; it is distinct from detecting emitted fluorescence or a reflected ultrasound pulse.

> **中文：** 光声成像用光激发吸收光的组织，并检测加热与热膨胀产生的声波。激发输入是光，测量输出是声；它区别于检测荧光发射或反射超声脉冲。

Source / 来源：Week 5 EE6301 Full Slides (AY2627).pdf, physical PDF p34–43

Prediction Q14: Answer C / 模拟Q14：答案C

EN: The slides associate charge transfer and a shared output node with CCD, and local pixel conversion/amplification with CMOS. Detector readout does not change the optical NA, and neither detector guarantees zero noise.

> **中文：** 课件将电荷转移及共享输出节点对应 CCD，将像素内转换/放大对应 CMOS。探测器读出方式不会改变光学 NA，两种探测器也都不保证零噪声。

Source / 来源：Week 4 EE6301 Full Slides (AY2627)(1).pdf, physical PDF p37–38

Prediction Q15: Answer A / 模拟Q15：答案A

EN: Amperometric sensing measures a reaction current at a selected potential. Potentiometric sensing instead measures potential with approximately negligible current. The use of an electrode alone does not distinguish the two; the measured quantity and operating condition do.

> **中文：** 电流型传感在选定电位下测反应电流。电位型则在电流近似可忽略时测电位。仅凭“使用电极”不能区分两者，应看被测量和工作条件。

Source / 来源：Week 6 EE6301 Full Slides (AY2627)(1).pdf, physical PDF p19–21 and p49

Prediction Q16: Answer D / 模拟Q16：答案D

EN: The measured signal is potential, so this pH electrode is potentiometric. An ion-sensitive interface responds selectively, while the reference provides a stable comparison. No Nernst calculation or memorisation of a long chemical equation is needed to make this classification.

中文： 所测信号是电位，因此该 pH 电极属于电位型传感。离子敏感界面提供选择性响应，参考电极提供稳定比较。做这个分类不需要 Nernst 计算或背长化学方程。

Source / 来源：Week 6 EE6301 Full Slides (AY2627)(1).pdf, physical PDF p39–43

Prediction Q17: Answer B / 模拟Q17：答案B

EN: ECG (electrocardiography) records cardiac electrical activity; EEG (electroencephalography) records brain-related electrical activity; EMG (electromyography) records muscle electrical activity. Thus B has all three matches correct. A and C swap two sources; D swaps ECG and EMG; E misidentifies all three. Surface EEG reflects the collective activity of many neurons and should not be treated as a direct recording of a single neuron’ s action potential.

中文： ECG（心电记录）记录心脏电活动；EEG（脑电记录）记录与脑活动有关的电信号；EMG（肌电记录）记录肌肉电活动。因此B的三项对应全部正确。A和C分别交换了两种来源；D交换了ECG与EMG；E的三项均错误。表面EEG反映许多神经元的群体活动，不能当作对一个神经元动作电位的直接记录。

Source / 来源：Week 7 EE6301 Full Slides (AY2627).pdf，PDF p7、p43–44、p81–83；T7-0004–T7-0006、T7-0085–T7-0087。

Prediction Q18: Answer D / 模拟Q18：答案D

EN: Depolarization means that the membrane potential becomes less negative. In this model, opening voltage-gated Na+ channels produces inward Na+ current during the rising phase. A describes the K+-associated return during repolarization; B describes hyperpolarization. C cannot explain the rising phase. E confuses the absence of an action potential with the absence of a resting membrane potential. The wording is limited to the neuronal model in the slides, rather than every type of excitable cell.

中文： Depolarization（去极化）指膜电位变得不那么负。在本模型中，电压门控Na+通道打开，Na+内流产生上升阶段。A描述复极化时与K+外流有关的恢复过程；B描述超极化；C不能解释上升阶段；E把“没有动作电位”误当作“没有静息膜电位”。题目限定课件中的神经元模型，没有把这一离子机制推广到所有可兴奋细胞。

Source / 来源：Week 7 EE6301 Full Slides (AY2627).pdf，PDF p21、p33–38；T7-0023–T7-0025、T7-0028–T7-0030。

Prediction Q19: Answer A / 模拟Q19：答案A

EN: Patch clamp uses a fine glass micropipette to form a membrane recording interface and study cellular ionic currents. The MEA diagrams show multiple electrode sites for spatially distributed electrical recordings. B assigns unrelated physical signals to the methods; C reverses their structures. Multiple MEA sites do not guarantee an independent intracellular measurement from every cell, so D is too strong. E confuses recording interfaces with signal-conditioning filters.

中文： 膜片钳使用细玻璃微吸管建立细胞膜记录界面，研究细胞离子电流。MEA示意图展示多个电极位点，可在不同空间位置记录电活动。B把无关的物理信号分配给这两种方法；C交换了两者结构。多个MEA位点并不保证能分别测到每个细胞的胞内信号，因此D的“必然”过强。E混淆了记录界面与信号处理滤波器。

Source / 来源：Week 8 EE6301 Full Slides (AY2627).pdf，PDF p9–13；T8-0011–T8-0016。

Prediction Q20: Answer E / 模拟Q20：答案E

EN: Vd = V1 − V2 = (s1 + n) − (s2 + n) = s1 − s2. Thus an identical common-mode interference cancels in this ideal model. A adds the inputs, and B discards the desired difference. C ignores the algebraic cancellation. D incorrectly extends ideal common-mode rejection to unequal interference and all real disturbances. Practical amplifiers and electrode imbalances can leave residual interference; no numerical CMRR formula is required here.

中文： Vd = V1 − V2 = (s1 + n) − (s2 + n) = s1 − s2。因此在此理想模型中，完全相同的共模干扰被抵消。A将输入相加；B丢掉了希望保留的信号差；C忽视了代数抵消；D错误地把理想共模抑制推广到不相同的干扰及一切真实扰动。实际放大器和电极不平衡可能留下残余干扰；此题不需要任何CMRR数值公式。

Source / 来源：Week 7 EE6301 Full Slides (AY2627).pdf，PDF p52–54；T7-0043–T7-0046。

Prediction Q21: Answer C / 模拟Q21：答案C

EN: High input impedance reduces current drawn from the electrode source and therefore reduces loading. Low output impedance helps the amplifier drive an external load with a smaller output voltage drop. Only C satisfies both requirements. A, B and E load the source strongly; D has the desired input property but an unsuitable output property. High input impedance alone does not determine voltage gain: gain is a separate amplifier characteristic.

中文： 高输入阻抗减少从电极信号源汲取的电流，从而减小加载效应。低输出阻抗有助于驱动外部负载，并减小输出端的电压降。只有C同时满足两项要求。A、B与E会明显加载信号源；D的输入特性合适，但输出特性不合适。高输入阻抗本身不决定电压增益，增益是放大器的另一项特性。

Source / 来源：Week 8 EE6301 Full Slides (AY2627).pdf，PDF p19–20；T8-0021–T8-0023。

Prediction Q22: Answer B / 模拟Q22：答案B

EN: A band-pass filter passes an intermediate band and attenuates lower and higher frequencies. A low-pass filter passes the low-frequency side; a high-pass filter passes the high-frequency side. A band-reject filter suppresses a selected intermediate band. An all-pass filter does not select a passband through magnitude attenuation. The 10–100 Hz band is specified only for this exercise and is not a universal ECG, EEG or EMG bandwidth. No RC cutoff calculation is involved.

中文： 带通滤波器通过中间频带，并衰减更低和更高的频率。低通滤波器通过低频一侧；高通滤波器通过高频一侧；带阻滤波器抑制选定的中间频带；全通滤波器不通过幅度衰减来选择通带。10–100 Hz仅为本题给定频带，不是ECG、EEG或EMG的统一频带。此题不涉及RC截止频率计算。

Source / 来源：Week 8 EE6301 Full Slides (AY2627).pdf，PDF p25–26；T8-0026–T8-0028。

Prediction Q23: Answer D / 模拟Q23：答案D

EN: First calculate the differential input: 1.30 − 1.10 = 0.20 mV. Then Vout = 1000 × 0.20 mV = 200 mV = 0.20 V. A omits the gain; B amplifies the mean input instead of the difference; C doubles the correct difference; E amplifies the input sum. The given gain is dimensionless and the output is positive because V1 is greater than V2.

中文： 先求差分输入：1.30 − 1.10 = 0.20 mV。再计算Vout = 1000 × 0.20 mV = 200 mV = 0.20 V。A漏掉增益；B放大的是输入均值而不是差值；C把正确差值又乘了2；E放大的是输入之和。给定增益没有单位；由于V1大于V2，输出为正。

Source / 来源：Week 7 EE6301 Full Slides (AY2627).pdf，PDF p53–54；T7-0044–T7-0046。

Prediction Q24: Answer C / 模拟Q24：答案C

EN: Series resistances add: Rtotal = 2.0 + 3.0 = 5.0 kΩ = 5000 Ω. By I = V/R, I = 0.010 V / 5000 Ω = 2.0 × 10^−6 A = 2.0 µA. A makes a factor-of-1000 unit error; B and D use only one resistance; E uses the difference of the resistances. This is an explicitly simplified pure-resistance model. Real electrode interfaces can include capacitance and half-cell potentials, so this model must not replace the full interface model in other questions.

中文： 串联电阻相加：Rtotal = 2.0 + 3.0 = 5.0 kΩ = 5000 Ω。由I = V/R，I = 0.010 V / 5000 Ω = 2.0 × 10^−6 A = 2.0 µA。A发生1000倍的单位错误；B与D只用了其中一个电阻；E使用了两电阻的差。本题明确采用简化纯电阻模型。真实电极界面还可能包含电容与半电池电位，因此不能在其他题中用本模型取代完整界面模型。

Source / 来源：Week 7 EE6301 Full Slides (AY2627).pdf，PDF p13、p41–42；T7-0007–T7-0008、T7-0035–T7-0038。

Prediction Q25: Answer B / 模拟Q25：答案B

EN: Use only the stated linear region: m = (700 − 100)/(80 − 20) = 10 pF per virus. The continuous-model LOD is 3 × 14 / 10 = 4.2 viruses. Because this question explicitly requires a whole-virus count at or above 4.2, take the ceiling: 5. Four is below the threshold and cannot be selected merely because it is the nearest integer. E is the correct continuous-model value but does not satisfy the integer requirement. This matches the teacher’ s W7 correction of sample Q10 to B/5. Do not automatically round all concentration-based LOD results upward; that step depends on the stated discrete-count requirement. The explicit assumption that the slope remains valid near the threshold avoids inferring low-count linearity merely from the two calibration points.

中文： 只使用题目指定的线性区：m = (700 − 100)/(80 − 20) = 10 pF/病毒。连续模型的LOD = 3 × 14 / 10 = 4.2个病毒。由于本题明确要求不低于4.2的完整病毒计数，应向上取整为5。4低于阈值，不能因为它是最近的整数就选4。E是正确的连续模型值，但不满足整数要求。这与老师W7对样题Q10的B/5更正一致。不要自动对所有按浓度表示的LOD都向上取整；这一步取决于题目是否规定离散整数计数。 题目明确假定该斜率在阈值附近有效，避免仅凭两个校准点推断低计数区仍线性。

Source / 来源：Sample Quiz-- For Students AY2627S1 (student).pdf，PDF p2 Q10；Week 1 EE6301 Full Slides (AY27).pdf，PDF p60–62；T7-0057–T7-0060。